

Table 2. Linear regression analysis reveals sex-specific genetic effects				
Model progression from baseline to key biological finding				
Model	Predictors	β^1	Model Fit	Key Finding
Model 1: Baseline			$R^2 = 0.019$	No genetic effect alone
	HI	-0.15 (0.67)		
	He	-0.39 (2.04)	$F = 1.94, p = 0.122$	
Model 2: Main effects			$R^2 = 0.063$	Infection dominates
	Infection	1.12 (0.29)***		
	Sex (Male)	-0.09 (0.28)		
	HI	0.19 (0.41)		
	He	1.56 (1.01)	$F = 5.05, p < 0.001$	
Model 3: Key interactions			$R^2 = 0.093$	Males deteriorate with
	Infection	2.32 (0.80)**		
	Sex × HI	2.14 (0.83)*		M. m. musculus ancestry
	Sex × Infection	-0.37 (0.57)	$F = 2.99, p = 0.001$	

¹ Significance: ***p < 0.001, **p < 0.01, *p < 0.05

Selected model (Model 3) shows males increase predicted weight loss by 2.14% per unit increase in M. m. musculus ancestry